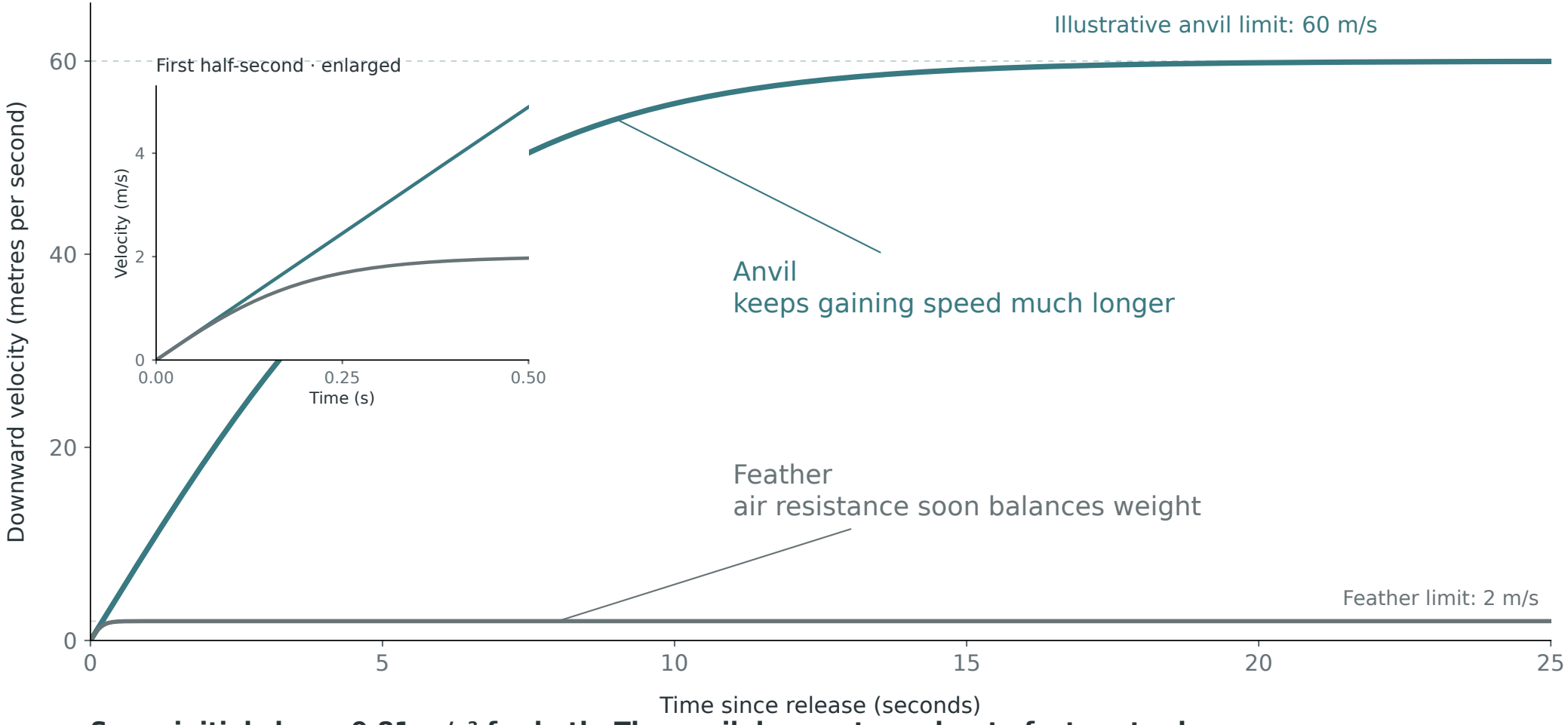
A feather and an anvil falling through air

Both start from rest with the same acceleration. Air resistance changes what happens next.



Same initial slope: 9.81 m/s² for both. The anvil does not accelerate faster at release.

Terminal speed means no further acceleration — not that the object stops falling.

Illustrative model, not measurements: chosen limits of 2 and 60 m/s. Real limits depend on mass, shape and air.

Enough falling distance is assumed; impact with the ground is not modeled.

Model: v(t) = V × tanh(9.81t / V); v = downward velocity, t = seconds, V = chosen terminal speed.

Physics reference: OpenStax, College Physics 2e, §5.2 “Drag Forces” · openstax.org/books/college-physics-2e/pages/5-2-drag-forces